Raymond Huang

Atlanta, GA | (201) 932-9488 | raymond.tingrui.huang@gmail.com | U.S. Citizen

Education

Georgia Institute of Technology — Atlanta, GA
B.S. in Electrical Engineering — *September 2026 – Expected 2028*

Boston University — Boston, MA
Electrical Engineering coursework; transferred to Georgia Tech | GPA 3.97 | Dean's List — *September 2025 – May 2026*

Skills

Programming: C++, Python, C, MATLAB, Verilog, Java, JavaScript
Robotics and Vision: ROS/ROS 2, PX4, Gazebo, Isaac Sim, OpenCV, PyTorch, YOLO
Hardware: ESP32, Arduino, Raspberry Pi, FPGAs, logic analyzer
Tools: Linux (Ubuntu), Git, GitHub, Fusion 360, AutoCAD

Experience

Undergraduate Researcher — September 2026 – Present
Georgia Institute of Technology, Laboratory for Intelligent Decision and Autonomous Robots (LIDAR) — *Atlanta, GA*

- Contribute to humanoid whole-body manipulation research by building on the lab's tactile sensing platform, which pairs wearable tactile sensors with a learned, force-aware planner to control a Unitree G1 humanoid.

Research Intern | Second author on paper submitted to ICRA 2027 — May 2026 – August 2026
Stevens Institute of Technology, Robotics and Automation Laboratory — *Hoboken, NJ*

- Made visuomotor policies robust to camera motion by expanding training data with novel-view synthesis from a feed-forward 3D Gaussian Splatting model; full method reached 97.5% success vs. 80% baseline over 40 trials.
- Trained a custom U-Net segmentation model in PyTorch on ∼500 hand-labeled images to mask out the robot arm during deployment, helping close the embodiment gap between human demonstrations and the robot.
- Calibrated and hardware-synchronized event and RGB-D cameras, and automated the data collection pipeline with Python scripts, cutting processing time per task from ∼8 hours to ∼30 minutes.
- Built a Gaussian Splatting simulation environment that supported visuomotor policy testing before hardware deployment.

Teaching Assistant — January 2026 – May 2026
Boston University Physics Department — *Boston, MA*

- Led 3 weekly problem-solving sections for ∼75 of the ∼500 students in an introductory mechanics course.
- Co-developed an animated traveling-wave simulation with fellow Teaching Assistants to teach wave concepts; across 208 pre/post survey responses, correct-answer rates rose from 51% to 83%.

Research Intern | Presented poster at 2024 IEEE MIT URTC — June 2023 – August 2023
Stevens Institute of Technology, Safe Autonomous Systems Lab — *Hoboken, NJ*

- Built a Gazebo simulation testbed with PX4 software-in-the-loop, a camera-equipped drone, and a moving ground robot.
- Fine-tuned YOLO to detect the robot in the drone's video feed; used camera geometry to estimate its 3D position.
- Developed an end-to-end C++ ROS pipeline combining detection, position estimation, and control; implemented and tuned PID controllers to send setpoints over MAVROS, enabling the drone to smoothly follow the ground robot at a set distance.

Projects

PetVision: Smart Multi-Pet Feeder | *1st Place, IEEE SSCS Arduino Contest* — June 2026 – August 2026

- Designed and built a pet feeder combining ESP32 control, camera-based pet identification, load-cell feedback, stepper-driven dispensing, and 3D-printed mechanical parts to manage individual feeding schedules and diets.
- Wrote C++ FreeRTOS firmware running network, command, and control tasks across both ESP32 cores to coordinate sensor readings (I2C load cell, SPI camera), dispensing, and app commands.
- Developed a YOLO and DINOv2 image-embedding pipeline to identify distinct pets (correctly identified pets in 40/40 self-collected test photos), and a React/Firebase mobile web app that communicates with the ESP32.

FPGA Reaction Timer | *Introduction to Logic Design course project* — March 2026 – May 2026

- Built a Verilog FPGA reaction timer with a state machine, random delay, and VGA output; tested with 10 testbench cases.